- 1 pair of axostyles, running along the midline
- Two sausage – shaped parabasal or median bodies lying transversely posterior to the sucking disc
- The trophozoite is motile, with a slow oscillation about its long axis, often resembling falling leaf (Figure 8.6a).

Cyst

It is the infective form of the parasite. The cyst is small and oval, measuring 12 µm × 8 µm and is surrounded by a hyaline cyst wall.

Its internal structure includes 2 pairs of nuclei grouped at one end. A young cyst contains 1 pair of nuclei. The axostyle lies diagnonally, forming a dividing line within cyst wall (Figure 8.6b).

8.3.4 Life Cycle: Giardia Passes Life Cycle in one Host, the Human

Infective form: Mature cyst

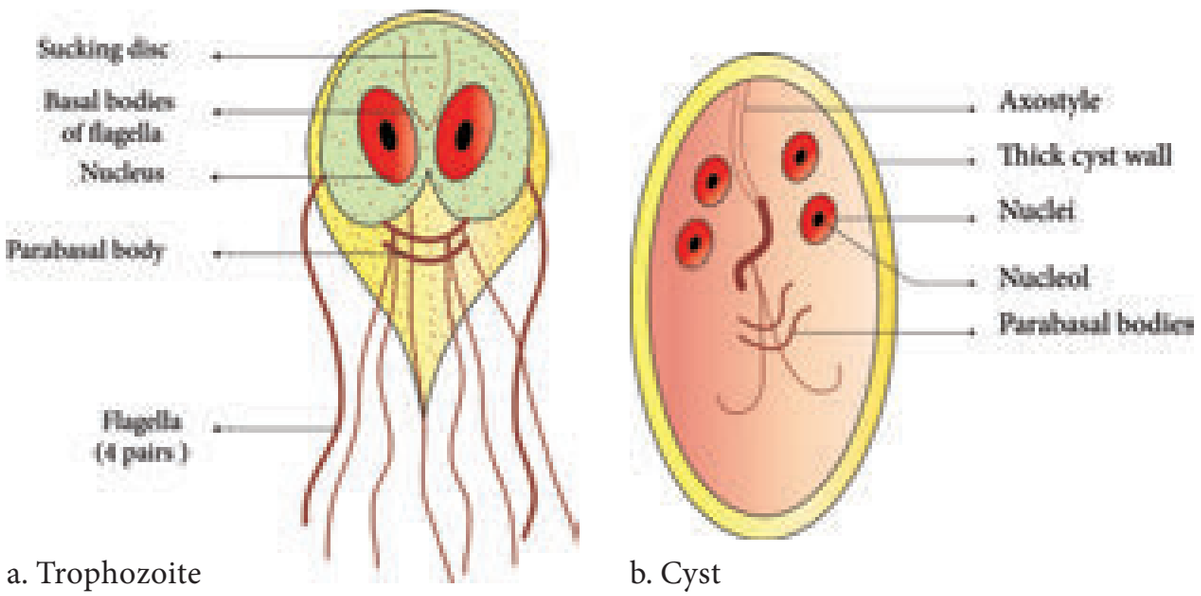
Mode of transmission: Human acquires infection by ingestion of cyst in contaminated water and food. Direct person – to person transmission occurs in children. Transmission occurs through oral-anal and oral-genital route in sexually active homosexual males. Within half an hour of ingestion, the cyst hatches out into two trophozoites, which multiply by binary fission and colonize in the duodenum. The trophozoites live in the duodenum and upper part of jejunum, feeding by pinocytosis. When conditions in duodenum are unfavourable, encystment occurs, usually in large intestine. Cysts are passed in stool and remain viable in soil and water for several weeks (Figure 8.7).

8.3.5 Pathogenicity

Giardia lamblia does not invade the tissue, but remains attached to intestinal epithelium by means of the sucking disc. It causes a disturbance of intestinal function leading to malabsorption of fat.

8.3.6 Clinical Manifestations

Incubation period is variable, but is usually about 2 weeks.



a. Trophozoite b. Cyst

Figure 8.6: Trophozoite and cyst of *Giardia*

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